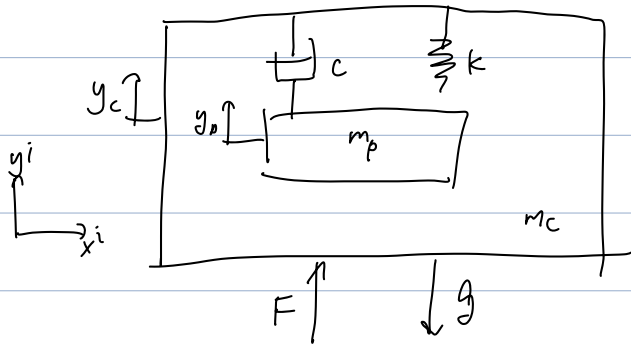
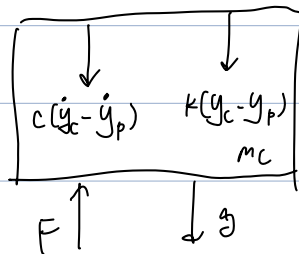


acc governing equation.



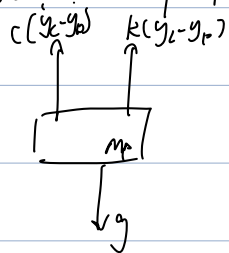
we begin by considering the case i



Applying Newton's second law,

$$\sum F = ma \Rightarrow m_c \ddot{y}_c = F - m_c g - c(\dot{y}_c - \dot{y}_p) - k(y_c - y_p) \quad (1)$$

Now, let's consider the proof mass:



Similarly, $m_p \ddot{y}_p = c(\dot{y}_c - \dot{y}_p) + k(y_c - y_p) - m_p g \quad (2)$

Subtracting ① and ②:

$$\ddot{y}_c - \ddot{y}_p = \frac{F}{m_c} - \left(\frac{1}{m_c} + \frac{1}{m_p}\right) \cdot c(\dot{y}_c - \dot{y}_p) - \left(\frac{1}{m_c} + \frac{1}{m_p}\right) \cdot k(y_c - y_p)$$

Let $\tilde{y} = y_c - y_p$, then

$$\ddot{\tilde{y}} = \frac{F}{m_c} - \left(\frac{1}{m_c} + \frac{1}{m_p}\right) \cdot c \cdot \dot{\tilde{y}} - \left(\frac{1}{m_c} + \frac{1}{m_p}\right) \cdot k \cdot \tilde{y}$$

giving:

$$\ddot{\tilde{y}} + c\dot{\tilde{y}}\left(\frac{1}{m_c} + \frac{1}{m_p}\right) + k\tilde{y}\left(\frac{1}{m_c} + \frac{1}{m_p}\right) = \frac{F}{m_c}$$

As $F = (m_c + m_p) a_{\text{input}}$

$$\Rightarrow \ddot{\tilde{y}} + c\dot{\tilde{y}}\left(\frac{1}{m_c} + \frac{1}{m_p}\right) + k\tilde{y}\left(\frac{1}{m_c} + \frac{1}{m_p}\right) = \left(1 + \frac{m_p}{m_c}\right) a_{\text{input}}$$

If $m_p \ll m_c$, then

$$\ddot{\tilde{y}} + \frac{c}{m_p} \dot{\tilde{y}} + \frac{k}{m_p} \tilde{y} = a_{\text{input}}$$

the steady-state solution is:

$$a_{\text{input}} = \frac{k}{m_p} \tilde{y}_{ss}$$